

Seungyeon Yoo

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EDUCATION

SEOUL NATIONAL UNIVERSITY

Seoul, Korea | Sep. 2021 – Feb. 2027

Ph.D. Candidate in Aerospace Engineering (GPA 4.04/4.3)

- Advisor: H. Jin Kim
- Research Interests: Robot Learning, Vision-Based Navigation, Visual Representation Learning
- Scholarship: BK Excellent Research Talent Fellowship

KOREA UNIVERSITY

Seoul, Korea | Mar. 2014 – Feb. 2020

B.S. in Mechanical Engineering (GPA 4.33/4.5)

- Scholarship: National Scholarship for Science and Engineering (merit-based; awarded for all 8 semesters)
- Awards: Grand Prize, Capstone Design Competition (₩2.0M, 2017); Academic Excellence Award (2017)

PUBLICATIONS

(*) Indicates equal contribution

- [1] **Seungyeon Yoo**, Youngseok Jang, Dabin Kim, Youngsoo Han, Seungwoo Jung, and H. Jin Kim, "ReaDy-Go: Real-to-Sim Dynamic 3D Gaussian Splatting Simulation for Environment-Specific Visual Navigation with Moving Obstacles," *IEEE Robotics and Automation Letters (RA-L)*, 2026.
- [2] Jusuk Lee, Daesol Cho, Jonghun Shin, **Seungyeon Yoo**, Jonghae Park, Taekbeon Lee, and H. Jin Kim, "BooST: Bridging Semantics and Motions for Efficient Skill Transfer," *IEEE Robotics and Automation Letters (RA-L)*, 2026.
- [3] Daesol Cho*, **Seungyeon Yoo***, Dongseok Shim, and H. Jin Kim, "Single-View 3D-Aware Representations for Reinforcement Learning by Cross-View Neural Radiance Fields," *IEEE Robotics and Automation Letters (RA-L)*, 2025.
- [4] Youngsoo Han, Youngseok Jang, Changhyeon Kim, **Seungyeon Yoo**, and H. Jin Kim, "Plane-Based Stereo Visual Localization with a Prior LiDAR Map," *IEEE Transactions on Intelligent Transportation Systems (T-ITS)*, 2025.
- [5] **Seungyeon Yoo***, Seungwoo Jung*, Yunwoo Lee, Dongseok Shim, and H. Jin Kim, "Mono-Camera-Only Target Chasing for a Drone in a Dense Environment by Cross-Modal Learning," *IEEE Robotics and Automation Letters (RA-L)*, 2024.
- [6] Seungwoo Jung*, Sangjin Han*, **Seungyeon Yoo***, H. Jin Kim, Sebastian Scherer, and Yunwoo Lee, "GAR-Tracker: Target-Visible Trajectory Planning for Ground Assistant Robots." (Under review)

PROFESSIONAL EXPERIENCES

SAMSUNG ADVANCED INSTITUTE OF TECHNOLOGY

Research Intern (Computer Vision Technology Unit at the AI Research Center)

Suwon, Korea | Jul. 2023 – Sep. 2023

SAMSUNG ELECTRONICS

Engineer (Manufacturing Core Technology Team at the Global Technology Center)

Suwon, Korea | Mar. 2020 – Jul. 2021

ILSA (DRIVER ASSISTANCE SYSTEM STARTUP TEAM)

Team Leader

Seoul, Korea | Jan. 2018 – Jun. 2019

PROJECTS

REAL-TO-SIM SIMULATION FOR HUMANOID LOCO-MANIPULATION

In collaboration with NVIDIA Research

Jun. 2026 – Present

AUTONOMOUS AND SWARM SYSTEMS USING VISION-AIDED NAVIGATION IN GNSS-DENIED AND JAMMING ENVIRONMENTS

Funded by Defense Acquisition Program Administration

Jan. 2025 – Present

LANGUAGE MODEL-DRIVEN INTERACTION AND SPATIAL LEARNING FOR PERSISTENT AUTONOMOUS ROBOT NAVIGATION

Funded by Samsung Science and Technology Foundation

Jan. 2025 – Present

HUMANOID ROBOT DEVELOPMENT FOR HUMAN-LIKE PERCEPTION, COMMUNICATION, AND LEARNING

Funded by Korea Ministry of Trade, Industry and Energy

Jan. 2025 – Dec. 2025

LEARNING-BASED ROAD SURFACE CONDITION ESTIMATION FOR VEHICLE SLIP SUPPRESSION

Funded by Hyundai Motor Company

Jan. 2023 – Dec. 2024

AUTONOMOUS DRIVING—ASSISTIVE ROBOT FOR FOLDABLE WHEELCHAIRS

Funded by Korea Ministry of Science and ICT

Aug. 2021 – Dec. 2022

EXTRACURRICULAR ACTIVITIES

INTERNATIONAL SPACE EDUCATION BOARD (ISEB)

Korea Representative Student

Baku, Azerbaijan | Oct. 2023

KOREA UNIVERSITY INTERNATIONAL STUDENT ASSISTANT (KUISA)

Personnel Leader / Group Leader

Seoul, Korea | Feb. 2017 – Dec. 2017

OTHER INFORMATION

KANGWON SCIENCE HIGH SCHOOL (Government-designated selective program for gifted students in science and engineering)

MILITARY SERVICE: Sergeant, Republic of Korea Army, Mar. 2015 – Dec. 2016